RED SEAL EXAM PREP

310S Automotive Service Technician

Free 50-Question Mock Exam

Practice for the Red Seal / Certificate of Qualification exam

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Mock Exam — answer all questions, then check the key

Q1. **When diagnosing a CAN bus communication fault (U-codes), what does a "Bus Off" condition indicate?**

A) A module counted too many errors and left the bus

B) Outdated scan tool software losing communication

C) The vehicle's power mode is in the accessory-off state

D) The CAN wiring has been disconnected from the battery

Q2. **A hybrid vehicle's high voltage battery has a state of charge (SOC) that drops rapidly under light load. The vehicle enters limp mode frequently. The HV battery pack voltage is within range at rest but drops significantly under load. What is the likely diagnosis?**

A) The drive motor consuming more power than it normally should

B) The onboard charger (OBC) not charging the battery while driving

C) Degraded cells or cell groups within the HV battery pack

D) A failing DC/DC converter drawing excess power from the pack

Q3. **A technician is using a test light to diagnose a circuit and finds voltage on both sides of a switch when the switch is open. What does this indicate?**

A) The switch is functioning correctly

B) The battery is deeply discharged

C) The load is drawing too much current

D) A short to voltage or a shorted switch

Q4. **A serpentine drive belt that is cracked, frayed, or glazed should be:**

A) Replaced immediately before it fails in service

B) Allowed to run as long as only one rib is cracked

C) Monitored until it breaks to confirm replacement is needed

D) Treated with belt dressing to restore grip and extend life

Q5. **A front-wheel-drive vehicle makes a clunk from the front suspension only when it is driven over a bump while turning under power, such as pulling over the lip of a driveway. With the vehicle standing still, turning the steering lock to lock produces no noise, and steering effort and return to centre are normal. The most likely cause is:**

A) A loose upper strut mount bearing

B) Air ingestion in the rack from low power steering fluid

C) A worn shock absorber topping out

D) A worn outer CV joint with play

Q6. **A technician diagnoses a P0894 code: "Transmission Component Slipping." Line pressure tests are within specification. What area should be investigated next?**

A) The oil pump output, since correct pressure eliminates the pump

B) Clutch pack clearance, worn friction plates, or a weak solenoid

C) The flexplate/flywheel, as it is the most common cause of P0894

D) The engine ignition timing, as retarded timing overloads the unit

Q7. **A vehicle has an intermittent no-crank condition. The starter does not engage when the key is turned to start. Battery voltage, starter relay, and starter motor test good. What is the MOST COMMON remaining cause?**

A) High resistance in the positive battery cable while cranking

B) The PCM commanding the starter off — reprogram the PCM

C) An intermittent open in an interlock switch or connector

D) A failed ignition switch not powering the starter relay coil

Q8. **What does a MAP sensor measure, and how does that reading affect fuel injection?**

A) Oil pressure — engine bearing protection

B) Intake manifold pressure — engine load

C) Fuel rail pressure — injector opening

D) Exhaust back pressure — ignition timing

Q9. **With the engine off, a technician presses the brake pedal several times to use up the reserve vacuum, then holds the pedal down and starts the engine. On a vehicle with a sound vacuum booster and check valve, what should happen?**

A) The pedal should sink slightly under the foot

B) The pedal should stay exactly where it is

C) The pedal should pulse in time with the engine

D) The pedal should rise firmly against the foot

Q10. **Sulfuric acid from a lead-acid battery splashes into a technician's eye. According to CCOHS, for how long should the eye be flushed with lukewarm, gently flowing water?**

A) For at least 30 minutes

B) For at least 60 minutes

C) For at least 10 minutes

D) For at least 15 minutes

Q11. **A vehicle with rear drum brakes has excessive brake pedal travel. The fluid level is correct, the pedal is firm once it comes up, it does not sink under steady pressure, and bleeding produces no air. What is the MOST likely cause?**

A) The rear wheel cylinders are leaking past the cups

B) The master cylinder is bypassing internally

C) The front brake pads are worn near the limit

D) The rear shoe adjusters have stopped working

Q12. **A pickup with a 340 cm wheelbase has been lifted so that its ride height is outside the maker's specification, and it is inspected in Manitoba. Its frame height at the highest point is 85 cm, and a 5 cm body lift is fitted. What is the smallest front track width that meets the handbook's track-width ratio for this modified truck?**

A) 153 cm

B) 180 cm

C) 170 cm

D) 162 cm

Q13. **What is the purpose of the transmission range sensor (also called the neutral safety switch or inhibitor switch)?**

A) To signal the TCM and ECM which gear range is selected

B) To engage manual shift mode from the sport gate

C) To control shift timing from vehicle speed and throttle

D) To prevent downshifts that would over-rev the engine

Q14. **A gasoline engine has a misfire on cylinder 3 at idle that clears at high RPM. The ignition coil, spark plug, and injector have been verified as good. The MOST likely cause is:**

A) A small vacuum leak at cylinder 3

B) Fuel pressure too low at idle

C) Low compression on cylinder 3

D) EGR valve stuck open on cylinder 3 port

Q15. **When bleeding brakes on a vehicle with an ABS system that has had air introduced into the HCU (Hydraulic Control Unit), what additional step is required?**

A) Remove and bench-bleed the HCU before reinstallation

B) An ABS activation cycle with a scan tool to purge the HCU

C) Flush the entire system with fresh fluid three times first

D) No additional steps; conventional bleeding is sufficient

Q16. **During a cooling system pressure test, pressure drops steadily over 10 minutes but no external leaks are visible. The MOST likely cause is:**

A) A leaking coolant pump shaft seal

B) Thermostat stuck open allowing pressure to escape

C) A faulty radiator pressure cap

D) An internal leak — head gasket, head, or block

Q17. **What is brake fade and what are the two main types?**

A) Gradual loss of braking efficiency; types are lining fade (overheated friction material) and fluid fade (boiling brake fluid)

B) Gradual loss of braking efficiency; types are mechanical fade (worn pads and scored rotors) and hydraulic fade (low reservoir fluid level)

C) Brake noise from glazed friction material; types are squeal fade (high-frequency pad vibration) and groan fade (low-speed shudder when stopping)

D) Increased pedal travel caused by air in the lines; types are static fade (parked on a grade) and dynamic fade (only while the vehicle is moving)

Q18. **When diagnosing automatic transmission shift quality, what does "hunting" (frequent up-down shifting at a steady speed) indicate?**

A) A worn torque converter causing inefficient power transfer

B) A shift point calibration, TPS, or shift solenoid fault

C) A low fluid level preventing full clutch pack engagement

D) Worn friction plates slipping in the direct clutch pack

Q19. **A stall test is performed to the specified procedure on an automatic transmission: brakes firmly applied, a drive range selected, throttle held wide open only for the few seconds allowed. The engine speed reached is well ABOVE the specified stall speed. What does that indicate?**

A) The engine is not producing its full power output

B) The converter stator one-way clutch is not holding

C) The torque converter clutch is applying too early

D) A clutch pack or band inside the unit is slipping

Q20. **A crankshaft is being installed in a block whose thrust bearing is a flanged main bearing. According to MAHLE's engine bearing brochure, how are the bearing's thrust faces aligned before the main cap bolts are fully tightened?**

A) Cap tapped toward the rear; crankshaft forced forward

B) Cap tapped toward the front; crankshaft forced rearward

C) Cap tapped toward the front; crankshaft forced forward

D) Cap tapped toward the rear; crankshaft forced rearward

Q21. **During a wheel alignment the rear thrust angle exceeds the manufacturer's specification, and rear toe is not adjustable on this vehicle. What is the correct procedure?**

A) Repair the rear suspension, then set front toe to the thrust line

B) Set front camber to offset the thrust angle and re-check tracking

C) Record the reading only — thrust angle does not affect tracking

D) Adjust front caster until the vehicle stops pulling to one side

Q22. **A customer fitted LED bulbs to a car's turn signals, and the signals now flash noticeably faster than before. Manitoba's inspection handbook has a reject line for the turn-signal flash rate. What range does that line allow?**

A) 70 to 150 flashes per minute

B) 40 to 120 flashes per minute

C) 50 to 130 flashes per minute

D) 30 to 110 flashes per minute

Q23. **An engine oil analysis shows high levels of iron and aluminum. The MOST likely source is:**

A) High sulphur content in the fuel supply

B) Coolant contamination from head gasket failure

C) Accelerated wear of internal engine parts

D) Use of the wrong engine oil viscosity

Q24. **How does R-1234yf differ from R-134a in flammability, and what does that mean for servicing?**

A) Both are non-flammable, so recovery, recharge, and leak-test service procedures are identical for the two refrigerants

B) R-1234yf is A2L (mildly flammable), R-134a is A1 (non-flammable) — use A2L-rated equipment, no open flames

C) R-134a is the mildly flammable refrigerant while R-1234yf is completely non-flammable in all service conditions

D) R-1234yf is A3 (highly flammable like propane) and is therefore prohibited from being serviced inside an enclosed shop

Q25. **What does "KOEO" mean in the context of automotive diagnostics?**

A) Key On, Engine Off

B) Key On, Engine Running

C) Key Off, Engine Off

D) Knock On, Engine Open

Q26. **Which refrigerant has replaced R-134a in most new light vehicles, and why?**

A) R-22, because it is non-flammable and inexpensive

B) R-410A, because it is shared with home air-conditioning systems

C) R-1234yf, because it has a far lower global warming potential

D) R-12, because it has better cooling capacity

Q27. **An 8-speed automatic transmission is displaying harsh shifts and a fault code for "Transmission Control Module — solenoid circuit fault, shift solenoid E." What is the likely consequence if this solenoid fails permanently open?**

A) The engine torque is limited to idle when the fault is detected

B) The transmission enters limp-home mode, locked in one gear

C) The transmission locks out all forward gears — reverse only

D) Only the commanded gear is affected — other gears work normally

Q28. **A vehicle pulls consistently to the left on a flat, level road with no crown. The two front tires are swapped side to side and the pull stays on the left. The front tires are then moved to the rear and the pull is still on the left. Both front rotors are the same temperature after the drive and both front wheels spin freely when raised. What caused the original pull?**

A) A defective (conical) tire causing the pull

B) A loose left front wheel bearing with play

C) Caster or camber off at the left front

D) Brake drag at the left front caliper

Q29. **A light-duty pickup inspected in Manitoba has a ten-leaf rear spring pack, and its main leaf is intact. Which finding does the handbook class as a hazardous condition, meaning the truck should not be driven, rather than only a general reject?**

A) Leaves shifted so they may touch a tire

B) One leaf cracked near the centre bolt

C) Two of its ten leaves broken through

D) A shackle bolt hole elongated

Q30. **During a Manitoba safety inspection, a car's daytime running lamps stay off with the engine idling, the transmission in Park, the parking brake applied and the headlamp switch off. What should the inspector do next?**

A) Retest in Park after the car has been driven several metres

B) Retest in Neutral with the parking brake released

C) Retest in forward gear with the parking brake released

D) Reject the DRLs as inoperative with the parking brake set

Q31. **A pickup is in for an Ontario Safety Standards Certificate. A pair of auxiliary high-beam lamps is bolted securely to its front bumper, but the owner has taken their bulbs out, so they cannot light. Every required lamp operates correctly. Under Ontario's standard, how do the auxiliary lamps affect the certificate?**

A) They are recorded as a defect, but do not fail it

B) They fail it until the bulbs are refitted and work

C) They are not judged until they are made to work

D) They fail it until both are taken off the truck

Q32. **A vehicle keeps bouncing two or three times after one end is pushed down and released, and the front tires show cupped, scalloped patches around the tread. What is the MOST LIKELY cause?**

A) Front wheel alignment toe set outside specification

B) Worn shock absorbers no longer damping the suspension

C) Front wheels out of balance, felt at highway speed

D) Sagging coil springs that have lost their free height

Q33. **An engine has a stretched timing chain. How would this most likely affect cam timing and what symptom would it produce?**

A) No effect on timing — the hydraulic tensioner takes up the slack and fully compensates for chain stretch

B) Cam timing becomes erratic — the only symptom is a ticking noise at idle

C) Cam timing advances — causes high idle, surging, and spark knock under load

D) Cam timing retards — causes rough idle, low power, and may set VVT fault codes

Q34. **A vehicle has excessive negative camber (top of tire tilts inward) on the front right wheel after a curb strike. What is the most likely damaged component?**

A) A worn wheel bearing with excessive play

B) A broken anti-roll bar link allowing suspension droop

C) A collapsed coil spring on the affected corner

D) A bent strut, lower control arm, or steering knuckle

Q35. **A connecting rod bearing removed from an engine has a cracked surface, with small irregular areas of lining material flaked out. According to MAHLE's bearing failure brochure, what damaged it?**

A) Oil starvation letting the journal wipe it

B) Hard particles embedded in the soft lining

C) Acids in the oil chemically etching it

D) Overloading from lugging or detonation

Q36. **A mass air flow (MAF) sensor reads lower than the airflow calculated from RPM and volumetric efficiency at every engine speed, and the gap between the two grows as airflow increases. Long-term fuel trim is only slightly positive at idle but climbs steeply under highway load. The MOST likely cause is:**

A) Air filter restriction causing low airflow

B) Air entering the intake after the MAF sensor

C) MAF sensor contaminated with oil deposits

D) Throttle body dirty causing incorrect air metering

Q37. **A manifold absolute pressure (MAP) sensor reads consistently HIGHER than actual barometric pressure at key-on, engine off. This MOST likely indicates:**

A) A throttle body sticking closed at idle

B) A vacuum leak downstream of the throttle

C) A failed MAP sensor or its signal circuit

D) A plugged or kinked MAP sensor vacuum line

Q38. **An ABS system stores a right rear wheel speed sensor fault code. The sensor resistance is within spec and voltage is present. The vehicle still activates ABS on every stop. The NEXT diagnostic step is:**

A) Check the brake pad thickness on that corner

B) Replace the wheel bearing assembly

C) Replace the ABS hydraulic control module

D) Inspect the tone ring for damaged or caked teeth

Q39. **What brake fluid specification is required for most vehicles equipped with ABS and electronic stability control?**

A) Glycol-based DOT 3, DOT 4, or DOT 5.1, as the OEM specifies

B) Mineral oil — ABS modulators require non-hygroscopic fluid

C) DOT 5 (silicone-based) — required for all ABS-equipped vehicles

D) Power steering fluid, since both are sealed hydraulic systems

Q40. **A relative compression test is performed with a lab scope and a current clamp on the battery cable while the engine is cranked with the ignition and injectors disabled. What is being compared between the cylinders?**

A) Starter current drawn as each cylinder compresses

B) Absolute cylinder pressure read from a screw-in gauge

C) Exhaust back pressure while the engine cranks

D) Fuel pressure held during the compression stroke

Q41. **What is the purpose of transmission fluid in an automatic transmission?**

A) It acts only as a coolant for the torque converter assembly

B) It lubricates, applies clutches, cools, and fills the converter

C) It only lubricates the moving parts to prevent internal wear

D) It provides only the hydraulic pressure needed for gear shifts

Q42. **A vehicle has a code P0171 (System Too Lean, Bank 1) and P0174 (System Too Lean, Bank 2). Both banks are lean. What is the MOST likely cause compared to a single-bank lean code?**

A) Both oxygen sensors failed simultaneously — replace both sensors

B) A shared fault: large vacuum leak, low fuel pressure, or weak MAF

C) A poisoned catalytic converter skewing downstream O2 readings

D) Low engine oil causing lean combustion through the crankcase vent

Q43. **What is "brake pedal pulsation" and what is the most common cause?**

A) Excessive pedal travel caused by badly worn brake pads

B) A soft pedal that gradually firms — caused by air in the lines

C) A braking vibration in the pedal from rotor thickness variation

D) Intermittent brake grabbing from a sticking caliper piston

Q44. **A brake-by-wire (BBW) system on a hybrid vehicle has no traditional brake booster. How is brake assist force generated?**

A) An electric motor-driven hydraulic pump builds pressure on demand

B) The driver simply applies more pedal force by design

C) The drive motor reverses to provide all of the braking force

D) An electric vacuum pump supplies conventional vacuum assist

Q45. **A dual-clutch transmission (DCT) hesitates and jerks during slow-speed manoeuvring in parking lots and traffic, and is normal at highway speeds. The owner reports it did not do this when the vehicle was new and that it has worsened steadily over the last two years. Fluid level and condition are correct and there are no leaks. What is the MOST LIKELY cause?**

A) Contaminated fluid stopping DCT operation at any speed

B) A failed speed sensor mis-selecting gears at low speed

C) Clutch engagement adaptation values drifting out of range

D) An inability of DCTs to operate at low speeds by design

Q46. **A vehicle's coolant temperature gauge reads normal, but the heater produces little heat at idle. When engine speed increases, heat output improves. What is the most likely cause?**

A) A stuck-open thermostat keeping the engine running too cold

B) A failed blend door actuator only partially opening at idle

C) A fully blocked heater core forced open at higher engine RPM

D) Low coolant level leaving air in the high-mounted heater core

Q47. **A light-duty pickup with rear leaf springs is in for an Ontario Safety Standards Certificate. The spring pin at one shackle is 19 mm in diameter, and no OEM wear figure is available. Vertical movement of the shackle against the pin is checked. Above what wear does Ontario's standard reject it?**

A) 1.0 mm

B) 3.0 mm

C) 2.0 mm

D) 4.0 mm

Q48. **During a brake fluid service, a technician finds that the brake fluid boiling point is significantly lower than specification. What is the MOST LIKELY cause and what is the risk?**

A) Wrong viscosity grade fluid with a lower boiling point

B) Contamination with engine oil from a master cylinder leak

C) Oxidation thickening the fluid and lowering its boiling point

D) Moisture absorbed by the hygroscopic fluid over time

Q49. **A failed A/C compressor is drained, and the oil that comes out is orange. According to MAHLE's thermal management damage brochure, what does that colour indicate?**

A) Metal wear particles carried in the oil

B) The compressor ran extremely hot inside

C) Too much UV leak-detection dye in the oil

D) Moisture reacting chemically with the oil

Q50. **A 2024 SUV built for the Canadian market has front fog lamps. In daylight, with the master light switch off, the driver turns the front fog lamps on in their normal steady-burning mode. Under section 108 of the Motor Vehicle Safety Regulations, which other lamps must then be on?**

A) No other lamps; the front fog lamps may operate on their own

B) Low-beam headlamps, with tail, licence plate and side marker lamps

C) Daytime running lamps, with tail, licence plate and side marker lamps

D) Parking lamps, with tail, licence plate and side marker lamps

Answer Key & Explanations

> Scoring: 35 correct out of 50 is 70%. These 50 questions come from the 183-question 310S bank; the last page shows what the complete printable bank contains.

Q1. Answer: A

"Bus Off" means a CAN module has counted too many transmit errors and taken itself off the network so it cannot disrupt the other modules. Error confinement is built into the protocol: each node keeps a transmit and a receive error counter. A node becomes error-passive once its transmit error counter passes 127, and it goes bus-off once that counter exceeds 255, at which point it stops transmitting altogether. Causes: shorts in the CAN wiring, a failing module, or severe electromagnetic interference. The remaining modules may still communicate, and the missing data from the silent module sets U-codes across the network. Find the root cause in the wiring or the module rather than clearing the codes and handing the vehicle back.

Q2. Answer: C

Rapid SOC drop with voltage sag under load: degraded HV battery cells. Li-ion/NiMH cells degrade over time — capacity reduces and internal resistance increases. Weak cells cause a rapid SOC drop and disproportionate voltage sag under discharge current. The BMS monitors individual cell voltages and temperatures; when cells deviate excessively, it limits power output (derate) or triggers limp mode/shutdown to protect the pack. Cell testing requires OEM-specific equipment.

Q3. Answer: D

Voltage on both sides of an open switch = unwanted path for voltage to reach the load side. Normally, with the switch open, the load side should show 0V (no path to positive). If voltage appears there, current is finding another path: either an internal short in the switch itself, or a short to voltage somewhere in the circuit between the switch and load. This can cause the component to operate even with the switch off. Diagnose by disconnecting suspect paths.

Q4. Answer: A

Damaged serpentine belt = replace immediately. One belt drives the alternator, the power steering pump, the A/C compressor and, on many engines, the water pump. A belt failure takes all of them out at the same moment: with no charging the vehicle runs the battery down and stops, and with no water pump it overheats quickly. A glazed belt slips and squeals under load and no longer drives those accessories properly. Belt dressing is not a repair — it masks slip noise, attracts dirt and can accelerate the breakdown of the belt.

Q5. Answer: D

The outer CV joint is the only one of these loaded by steering angle, suspension travel and drive torque at the same moment. As its cage and balls wear, the joint develops lash; put it at a large angle, compress the suspension and apply torque, and it takes up that lash with a clunk — the classic driveway-lip noise on a front-drive car. An upper strut mount bearing can clunk too, and on a MacPherson strut it is a common source of noise when turning; what rules it out here is the second half of the stem. A failing mount bearing makes its noise whenever the strut has to rotate, so it can be reproduced by steering the vehicle while it stands still, and it normally stiffens steering effort or slows the return to centre. Neither happens on this vehicle. A shock topping out clunks in rebound over a bump whether or not the vehicle is turning, and air drawn into the rack gives a groan or buzz that follows steering effort rather than bumps. Confirm by checking the axle for play at full lock and looking for a split boot with grease thrown around the inner wheel well.

Q6. Answer: B

P0894 (slipping) with normal line pressure: focus on clutch pack wear, clearance, or solenoid control of specific elements. Normal line pressure confirms the pump and main pressure regulator are working. Slipping must be occurring at a specific clutch pack or band. Causes: worn friction plates (excess clearance), weak apply pressure at a specific solenoid circuit (solenoid not fully applying that clutch), or worn clutch pack seals. Use a clutch application chart to identify which elements are in use during slipping conditions.

Q7. Answer: C

Intermittent no-crank is most often an intermittent open in the starter enable circuit. The starter is not commanded directly by the key: the request passes through a series of interlocks before it reaches the relay coil - the park/neutral position switch on an automatic, the clutch pedal switch on a manual, and on many vehicles an immobilizer or body module permission. Anything in series that opens even briefly prevents cranking, and corroded or loose connectors in that chain are the classic cause. The starter motor and relay have already tested good, and high cable resistance would give slow cranking rather than no engagement at all. Reproduce the fault before testing: wiggle the shift lever through Park and Neutral, hold the clutch at different pedal positions, and watch for voltage at the solenoid S terminal during a failed attempt.

Q8. Answer: B

MAP = engine load sensor. It measures absolute pressure in the intake manifold. High vacuum (low absolute pressure) means light load with little air entering, so the ECM commands a shorter injector pulse width — less fuel, not a leaner mixture. Low vacuum (high absolute pressure) means heavy load with more air entering, so pulse width increases. The ECM combines MAP with engine speed and intake air temperature to work out the mass of air entering the cylinders, then meters fuel to the target ratio. Oil pressure, fuel rail pressure and exhaust back pressure each have their own dedicated sensor.

Q9. Answer: A

The pedal drops because the booster has just been given the vacuum it needs. A vacuum booster works on the difference between atmospheric pressure on one side of its diaphragm and manifold vacuum on the other, and it uses that difference to add force to whatever the driver applies. Pumping the pedal with the engine off exhausts the reserve vacuum stored in the booster and held there by the one-way check valve, so the last applications feel hard and high - that is the unassisted pedal. Start the engine and vacuum reaches the booster, the diaphragm is pushed, and the assist appears under a foot that is already pressing, so the pedal sinks. A pedal that does not move means the booster is not receiving vacuum or cannot hold it: check the hose for splits or a poor seal, check the one-way check valve, and check the diaphragm. Nothing in this test should push the pedal back up, and a pulsation with the engine running is not part of it. Note what the test does not prove: it checks the assist, not the hydraulics, so a pedal that afterwards sinks slowly toward the floor under steady pressure is a master cylinder bypassing internally and a separate fault.

Q10. Answer: A

At least 30 minutes. CCOHS's Garages - Lead-acid Batteries guidance says to immediately flush the contaminated eye with lukewarm, gently flowing water for at least 30 minutes while holding the eyelid open, to repeat flushing if irritation persists, and not to interrupt flushing. It warns not to rinse contaminated water into the unaffected eye or onto the face, and to call for medical help and transport the victim to an emergency care facility. A 10- or 15-minute flush stops short of the time CCOHS gives, and 60 minutes is not the figure it gives for battery acid.

Q11. Answer: D

Excessive travel with a pedal that is firm and holds is a mechanical clearance problem, not a hydraulic one. Drum brakes need the shoes held close to the drum, and the automatic adjuster takes up lining wear a little at a time so the pedal stays high. When the adjuster seizes with rust, or its lever is bent, unhooked or reinstalled wrongly, the shoes stay where the last adjustment left them and the gap grows as the linings wear, so more and more pedal travel is spent moving the shoes out to the drum before any braking starts. Each of the other answers produces a different pedal. A master cylinder bypassing internally gives a pedal that sinks slowly under steady pressure and firms up when pumped - excluded by the stem. Leaking wheel cylinders lose fluid, so the reservoir level falls and the linings and backing plate are wet - also excluded. Worn front disc pads add almost no travel, because the caliper piston follows the pad outward as the pad wears. Pull the drums, free and lubricate the adjuster, verify the lever action, adjust the shoes to the specified drag and then check pedal height and the parking brake adjustment.

Q12. Answer: D

162 cm: 1.80 × (85 cm + 5 cm) = 162 cm. Manitoba's handbook treats a vehicle whose suspension has been altered so that its ride height is no longer within the maker's specification as a modified vehicle. For a modified truck, MPV or SUV with a wheelbase of 254 cm (100 in.) or more, it rejects the vehicle when the front track width divided by the sum of the frame height at the vehicle's highest point plus any body lift is less than 1.80; with a wheelbase under 254 cm the ratio is 2.00. Track width is the distance between the centre points of the left and right tire treads. Here the divisor is 85 + 5 = 90 cm, and 90 × 1.80 = 162 cm, so any narrower track fails. 153 cm leaves out the body lift and fails, while 170 cm and 180 cm come from the 2.00 short-wheelbase ratio and ask more than the handbook requires for this truck.

Q13. Answer: A

Range sensor / neutral safety switch: signals gear position, prevents starting in gear. Two critical functions: 1) Neutral safety — allows starter engagement only in Park or Neutral, preventing the engine from starting in gear. 2) Gear position signal — tells the ECM/TCM what range (P, R, N, D, etc.) is selected so it can adjust shift strategy, ignition timing, and emissions controls accordingly. Failure: no-start in P or N (safety switch aspect), incorrect gear position displayed, or shift quality issues.

Q14. Answer: A

Idle misfire only (clears at high RPM) = vacuum leak. Vacuum leaks have a greater dilution effect at idle (small throttle opening, high manifold vacuum). At high RPM, manifold vacuum drops and the additional air from the leak is a smaller percentage of total flow — misfire disappears. Typical single-cylinder leak points are the intake manifold runner gasket and the injector O-ring. Use propane or smoke to locate vacuum leaks.

Q15. Answer: B

Air trapped in HCU solenoid passages requires a scan tool to cycle ABS components for complete bleeding. Conventional bleeding clears the main hydraulic lines but cannot reach fluid trapped in the ABS modulator's small solenoid passages and accumulator. The scan tool activates each solenoid valve and the pump motor in sequence, moving air pockets to where they can be bled out. Each OEM specifies the sequence. Not performing this step leaves air in the HCU — spongey pedal returns.

Q16. Answer: D

Pressure loss without external leaks = internal leak. A failing head gasket, cracked head, or cracked block lets coolant leak into the engine oil (milky dipstick), the combustion chamber (white exhaust smoke), or between cylinders (bubbles in coolant). Use a combustion leak test kit (checks for exhaust hydrocarbons in coolant) to confirm head gasket failure.

Q17. Answer: A

Brake fade: reduced braking effectiveness from heat. Two types: lining fade and fluid/vapor lock fade. Lining fade: friction material overheats, its coefficient of friction drops, reducing braking force despite normal pedal pressure. Fluid fade (vapor lock): brake fluid boils, creating compressible gas bubbles — pedal goes low and spongy. Both occur during extended/severe braking (long downhill grades). Solutions: upgraded brake fluid (higher dry/wet boiling point), cross-drilled/slotted rotors, performance brake pads.

Q18. Answer: B

Shift hunting: the transmission cycles between two gears at a steady speed — a shift control problem, not a mechanical wear problem. Incorrect shift point calibration, a throttle position sensor fault, or a shift solenoid fault are the usual causes. The TCM uses vehicle speed, throttle position, engine load and gear ratio to decide shift points. If the throttle position signal is erratic, the TCM sees load conditions that keep alternating and commands one shift after another; an erratic vehicle speed signal does the same thing for the same reason, because both are shift-scheduling inputs. Worn friction material, by contrast, shows up as slipping or flare during a shift, not as repeated cycling between two gears. Check: throttle position voltage sweep (smooth?), the vehicle speed signal on a graphing meter, shift solenoid response, and fluid level and condition.

Q19. Answer: D

Stall speed is the point where the engine's torque output and the converter's ability to absorb it balance, so a reading above specification means torque is escaping behind the converter. With the vehicle held still the converter turbine cannot turn, and the engine winds up until the converter is absorbing everything the engine can produce. If a holding element inside the transmission - a clutch pack or a band for the range selected - cannot hold, the driveline gives way before the converter loads the engine fully, so the engine runs past its specified stall speed. A reading BELOW specification means the opposite: either the engine cannot make its rated torque, or the converter's stator one-way clutch is freewheeling instead of holding, so the converter never multiplies torque and the engine reaches its ceiling early. Compare the reading with the figure for that engine and transmission combination, repeat it in each drive range to see which ranges misbehave, and read the result alongside a line pressure test and a road test rather than on its own. The test is hard on the unit: hold it only for the few seconds the procedure allows, watch fluid temperature, and let the transmission cool between attempts.

Q20. Answer: A

MAHLE has the main cap tapped toward the rear of the engine, then the crankshaft forced as far forward as possible to align the rear thrust faces while the cap bolts are snugged. MAHLE's engine bearing brochure gives the sequence for assembling thrust bearings: tighten the main cap bolts to approximately 10 to 15 ft. lbs. to seat the bearings, then loosen them; tap the main cap toward the rear of the engine with a soft-faced hammer; tighten the cap bolts finger tight; using a bar, force the crankshaft as far forward in the block as possible to align the bearing's rear thrust faces; while holding the shaft in the forward position, tighten the cap bolts to 10 to 15 ft. lbs.; then complete the tightening to specification in 2 or 3 equal steps. MAHLE says this procedure should align the bearing thrust faces with the crankshaft to maximize the bearing area in contact for carrying load. Its installation guidelines likewise check end clearance with the crankshaft forced all the way forward in the block.

Q21. Answer: A

A rear axle out of thrust specification is a repair job, not a front-end adjustment. Thrust angle is the direction the rear axle points relative to the vehicle centreline, and it is set by rear toe. Out of specification, the vehicle dog-tracks and the steering wheel sits off centre. Nothing at the front corrects it: camber and caster do not change where the rear axle points, and dialling front toe alone only straightens the steering wheel while the vehicle still crabs down the road. With rear toe non-adjustable, inspect the rear end for a bent axle or trailing arm, a shifted cradle or subframe, collapsed or displaced bushings and collision damage, then reposition, shim or replace whatever is out of position — many non-adjustable rear suspensions accept offset bushings or shims for exactly this. Once the rear is back within specification, set front toe to the thrust line rather than to the geometric centreline.

Q22. Answer: C

Between 50 and 130 flashes per minute. Manitoba's Vehicle Inspection Handbook rejects turn signal lamps that do not flash between 50 and 130 times per minute, so a set that flashes faster than 130 a minute after an LED conversion fails on that line, even if every lamp lights. The same item also rejects turn signals that are not activated only by the turn signal switch, and a switch that does not remain in its selected position and activate only the proper lamps.

Q23. Answer: C

Iron + aluminum in oil = engine internal wear. Iron comes from cylinder walls, rings, and camshaft wear. Aluminum comes from pistons, bearings, and thrust washers. Elevated levels indicate faster-than-normal wear. Cross-reference with mileage and other elements to determine severity — coolant intrusion shows up as sodium, potassium, boron or glycol rather than as wear metals.

Q24. Answer: B

R-1234yf is classified A2L (mildly flammable); R-134a is A1 (non-flammable). A2L is the lower-flammability class. R-1234yf burns slowly — a burn velocity of about 1.5 cm/s against 46 cm/s for propane — and its heat of combustion is roughly a quarter of propane's. It also takes thousands of millijoules to ignite, against about 0.25 mJ for propane, so it is far harder to set alight than a hydrocarbon refrigerant. It will still burn, so servicing requires A2L-rated recovery/recharge equipment, adequate ventilation, a refrigerant-specific electronic leak detector, and no open flames or ignition sources near the work. Systems are also designed with features to manage the small flammability risk.

Q25. Answer: A

KOEO = Key On, Engine Off. Many sensor tests and circuit checks are performed KOEO — ignition key in ON position but engine not running. This powers the electrical system without the noise of the running engine. KOER = Key On, Engine Running (for oxygen sensor, fuel trim, etc.). Important to know which mode specific tests require.

Q26. Answer: C

R-1234yf (an HFO) replaced R-134a in new light vehicles because of its far lower global warming potential (GWP ~4 vs ~1430 for R-134a). R-1234yf is not interchangeable with R-134a: it uses unique service fittings, dedicated recovery/recharge equipment, and is classified A2L (mildly flammable). Never top up an R-1234yf system with R-134a or vice versa.

Q27. Answer: B

Shift solenoid fault triggers limp-home mode — transmission locks into a safe fixed gear. Modern transmissions with electrical faults default to limp-home (fail-safe) mode. The TCM locks the transmission in one gear (commonly 2nd or 3rd) to allow the vehicle to be driven safely to a service facility while preventing further damage. This prevents: multiple gear engagement simultaneously (mechanical damage), loss of all drive. The driver experiences loss of all shifting ability, engine RPM limited by the fixed gear, and difficulty accelerating from stop.

Q28. Answer: C

A pull that survives both moves is not coming from a tire — it belongs to the corner. Conicity and plysteer are built-in lateral forces: the tire pushes sideways in a fixed direction relative to itself, so moving it from one front position to the other inverts the direction of that push. The pull then reverses, or cancels against the other front tire, and once the tire is at the rear its lateral force has little effect on steering. Neither happened here, so the tires are ruled out. Equal rotor temperatures after the drive and wheels that spin freely rule out brake drag at the left front, and a loose wheel bearing shows as play at the wheel, noise and wander rather than a steady one-way pull. What is left is the alignment angles that generate side force at a steered wheel: the vehicle leads toward the side with the more positive camber and toward the side with the less positive caster, so the left front is the corner to measure. Put the vehicle on an alignment rack, check ride height first because a sagged spring moves both angles, then compare left and right caster and camber before any part is replaced. Radial force variation is a separate defect: it varies the tire's radial stiffness through each revolution and shows as vibration and shake, not as a pull.

Q29. Answer: A

Manitoba lists leaves displaced so that they could contact a tire, rim or brake component as a hazardous condition. The handbook's leaf-spring item puts these in its hazardous column: in any leaf spring assembly, the main leaf is broken or missing or at least 25% of the leaves are broken or missing; and one or more leaves displaced in a manner that could result in contact with a tire, rim or brake component. Two broken leaves out of ten is 20%, below the 25% threshold, so with the main leaf intact it is a general reject: the truck fails, but the handbook does not class it as too dangerous to drive. A cracked leaf and an elongated shackle bolt hole are general rejects too. The handbook defines a hazardous condition as one with safety implications so serious that the vehicle should not be driven.

Q30. Answer: C

Retest with the transmission in a forward gear and the parking brake off. Manitoba's Vehicle Inspection Handbook says daytime running lamps are inspected with the engine running, the transmission in forward gear, the parking brake off and the headlamp switch in the OFF position, and notes that vehicle design may require the vehicle to be driven several metres before the system activates. DRLs that stay off in Park with the parking brake set have not yet been tested under those conditions. Rejecting now skips the handbook's test conditions, and a retest in Neutral, or back in Park after the car has been driven several metres, still does not have the transmission in a forward gear with the parking brake off. If the lamps then do not operate continually, that is a reject.

Q31. Answer: C

Under Ontario's auxiliary-lamp item, lamps that cannot light are not judged until they are made to work. Ontario's Passenger / Light-Duty Vehicle Inspection Standard counts auxiliary low and high beam headlamps and front and rear fog lamps as auxiliary lamps. Its note says these lamps are not required, but if functional, must be inspected, and that all auxiliary lamps on the front or rear of the vehicle must comply with the auxiliary-lamp requirements if the lamps are operational. With their bulbs out these lamps are not operational, so those requirements do not apply to them yet, and the auxiliary-lamp item does not fail the vehicle. Nor are auxiliary lamps among the items the standard lists as recordable on the inspection report. Once the lamps are made to work, they must meet the auxiliary-lamp requirements, such as auxiliary headlamps that are white in colour and proper aim for the lamp type. Manitoba words its optional-lamp note differently: such lamps, if present on a vehicle, must meet its requirements.

Q32. Answer: B

A suspension that keeps oscillating has lost its damping, and cupped tire wear is the same fault written on the tread. The spring carries the weight and stores the energy of a bump; the shock absorber or strut converts that energy into heat so the spring stops after one movement instead of oscillating. A worn damper cannot do that, so the corner keeps bouncing after it is pushed and released, and the vehicle floats over dips and feels unsettled at speed. The tire pays for it: as the wheel bounces, the contact patch is alternately loaded and unloaded, and the tread wears into a pattern of high and low patches around its circumference - cupping, or scalloping. The other three answers each show themselves differently. Sagged springs change ride height and therefore the alignment angles, but they do not allow repeated oscillation. Out-of-balance wheels vibrate at a speed tied to wheel rotation and have no effect on the bounce test. Incorrect toe scrubs the tread with a feathered edge across the face of the tread, not in patches around it. Confirm by comparing the bounce at each corner, looking for oil wetting down the damper body, and checking the mounts and bushings before condemning the unit.

Q33. Answer: D

Stretched timing chain retards cam timing, reducing power and causing rough running. Chain stretch effectively lengthens the chain, causing the camshaft to lag behind the crankshaft — retarded cam timing. Symptoms: reduced power especially at low-end, rough idle, rattling noise on start-up (tensioner over-extended), and VVT system codes because the ECM cannot achieve target cam advance. Chain stretch beyond tensioner range = timing jumps possible (catastrophic).

Q34. Answer: D

Camber beyond adjustment range after impact: structural damage. Camber is primarily a function of suspension geometry — strut angle, control arm length, and knuckle design. These structural components control the camber angle; a hard impact can deform them, changing camber beyond the adjustment range permanently. Inspect with a straightedge and alignment rack. Replace bent components before alignment.

Q35. Answer: D

A cracked surface with small irregular pieces missing is surface fatigue, and overloading such as lugging or detonation is the first cause MAHLE lists for it. MAHLE's bearing failure brochure says heavy pulsating loads from the reciprocating motion within the engine crack the bearing surface through metal fatigue; the cracks deepen, then turn and run parallel to the bond line until pieces of the surface flake out. It lists overloading (lugging, detonation or overfueling), uneven loading, a bearing material of inadequate fatigue strength, and a normal life span that has simply been exceeded, and it advises replacing all the other bearings as well, since their remaining service life may be short. Oil starvation leaves a very shiny, wiped surface that turns dark blue to black, foreign particles are embedded in the lining with scratch marks, and corrosion leaves a darkened, spongy, etched surface.

Q36. Answer: C

Read the trim pattern, not just the low reading. A MAF element coated with oil — typically from an over-oiled reusable filter — or with fine dust is insulated, so it gives up its heat more slowly and under-reports airflow. The error scales with airflow: small at idle, large at load. The ECM fuels from that low reading, so the engine leans out most where the error is biggest, which is exactly the trim pattern described. Unmetered air entering after the sensor produces the opposite pattern, because the leak is a large fraction of total airflow at idle and a small fraction once the throttle opens — those vehicles show their worst trims at idle and clean up under load. A restricted air filter genuinely reduces airflow, and the MAF reports that reduced airflow honestly, so fuel still matches air and trims stay near zero. A dirty throttle body upsets idle air control and idle quality, not MAF accuracy. Compare grams per second at idle and at wide-open throttle against expected values, then clean the element with MAF-specific cleaner and recheck before condemning the sensor.

Q37. Answer: C

With the engine stopped there is no vacuum anywhere in the system, so no vacuum-related fault can be the answer. Key-on, engine off, the intake manifold sits open to atmosphere through the throttle and the MAP sensor should report barometric pressure — which falls with altitude, so a sea-level figure is not the standard everywhere. A plugged or kinked hose simply traps air at atmospheric pressure, which is the correct reading under this condition; a leak downstream of the throttle likewise changes nothing when no vacuum exists; and a throttle held closed cannot raise manifold pressure above atmosphere. A reading above barometric with the engine off can therefore only come from the sensor or its wiring: a sensor that has drifted high, a shorted or biased signal line, or a corroded connector raising signal voltage. Compare the MAP and BARO parameters on the scan tool at key-on — they should agree within a few kPa — then back-probe the sensor for reference voltage, ground and signal before replacing it. A blocked or disconnected vacuum line does show itself, but only with the engine running, where MAP stays near barometric instead of dropping.

Q38. Answer: D

Correct sensor but wrong signal = tone ring problem. Damaged, cracked, missing, or packed-with-debris tone ring (reluctor ring) teeth create an irregular signal that the ABS module interprets as wheel lockup. This triggers ABS unnecessarily. Clean the tone ring and inspect for damaged teeth. Common on vehicles that have driven through deep mud or water.

Q39. Answer: A

ABS/ESC vehicles: use the OEM-specified DOT rating, usually DOT 4 or DOT 5.1. DOT 3, DOT 4, DOT 4 LV and DOT 5.1 are all glycol-based and miscible with one another, so the system is filled and topped up with the grade the manufacturer calls out. DOT 5 is silicone-based and is not miscible with the glycol fluid these systems are filled with; because it does not absorb moisture, free water collects at low points where it corrodes components and can boil under hard braking, and silicone fluid aerates and compresses more readily under rapid ABS modulator cycling. ABS and ESC vehicles are commonly specified DOT 4, DOT 4 LV or DOT 5.1 for both a higher wet boiling point and a lower cold viscosity, so the modulator pump can move fluid quickly in winter. Never mix DOT 5 with DOT 3, DOT 4 or DOT 5.1, and confirm the specification on the reservoir cap or in the service information before adding fluid.

Q40. Answer: A

The scope-and-clamp version of this test uses the starter as the measuring instrument. Every compression stroke loads the starter, so cranking current rises and falls once per cylinder, and the height of each hump is proportional to how hard that cylinder is to compress. A cylinder with a leaking valve, a broken ring or a breached head gasket compresses less air, loads the starter less, and leaves a shallow hump in the waveform. Nothing is disassembled and no gauge is fitted, which makes it a fast screening test — but it is relative: it shows which cylinder is weak, not how many kPa that cylinder holds, so a low cylinder is confirmed with a screw-in compression gauge and then a leak-down test to find where the air is going. The scan tool version of the same test reaches the same conclusion from the other side of the mechanism, graphing crankshaft speed instead of current. Fuel pressure, exhaust back pressure and absolute cylinder pressure are each read with their own gauge and say nothing about compression balance between cylinders.

Q41. Answer: B

ATF does four jobs at once: lubrication, hydraulic pressure for clutch and band application, heat transfer, and fluid coupling in the torque converter. It lubricates planetary gearsets, bearings and bushings. It carries the hydraulic pressure that applies the clutch packs and bands to engage each gear. It picks up heat and carries it to the cooler. It fills the torque converter, where it couples the engine to the transmission and multiplies torque. Using a fluid with the wrong friction modifiers causes shift quality complaints, clutch slip and converter shudder.

Q42. Answer: B

Lean on both banks: look upstream (fuel system or air measurement), not at individual cylinders. A single-bank lean code (P0171 only) suggests a bank-specific issue (vacuum leak near that bank, bank injectors, O2 sensor). BOTH banks lean simultaneously points to a system-wide shared component: MAF sensor (undercounts airflow — ECM underfuels), fuel pump or clogged filter (low pressure affecting all injectors), a large vacuum leak after the MAF sensor, or all injectors under-delivering.

Q43. Answer: C

Pedal pulsation: rotor thickness variation (DTV). Pulsation is a vibration felt through the pedal during braking. A rotor that is not perfectly uniform in thickness around its circumference pushes the caliper pistons in and out as it rotates — this pressure variation is felt as a pulsation through the pedal. DTV is caused by excessive runout (rotor wobble) that lets the pads touch and wear the rotor unevenly, or by improper wheel torque (over-torquing warps rotors). Check: measure rotor thickness at 8+ points.

Q44. Answer: A

Brake-by-wire: electric hydraulic pump provides assist, blends with regen braking. Hybrid/EV brake systems use electric-motor-driven pumps (or electrohydraulic actuators) to provide brake pressure on demand without a conventional vacuum booster. The system must seamlessly blend regenerative braking (electric motor generates electricity while slowing) with friction brakes — prioritizing regen for energy recovery and using friction brakes to supplement, managing the transition smoothly. This requires sophisticated control.

Q45. Answer: C

A low-speed complaint that was absent when the vehicle was new and has grown steadily points at clutch wear moving the engagement point beyond what the TCM can adapt to. A DCT learns the point at which each clutch begins to transmit torque and stores it as an adaptation value. As the friction material wears, the real engagement point moves and the TCM shifts its adaptation to follow. Once the required correction runs past the end of the adaptation table, engagement comes too early or too late and the driver feels hesitation and jerking at crawling speeds, where the clutch is being slipped the most. At highway speed the clutch is fully engaged, so the complaint disappears — which is why the condition it appears under is the strongest clue. Read the adaptation values, then perform an adaptation reset and relearn; if the values run out of range again, the clutch is worn past its service limit. Note the contrast with a DCT that has jerked at low speed since new, with adaptation in range and a same-model vehicle behaving identically — that is the normal characteristic of clutch engagement, not wear.

Q46. Answer: D

Low coolant level: poor heater output at idle that improves with RPM is a classic low coolant symptom. When coolant level is low, there is insufficient coolant flow through the heater core (located high in the dash), which may be partially filled with air. At idle, low pump flow allows air pockets to remain in the heater core. At higher RPM, increased coolant pressure and flow forces coolant through the air pocket. Also check: coolant level drops over time (leaking head gasket, external leak). Air in system: bubbles, fluctuating temp gauge.

Q47. Answer: C

For a spring pin up to 25 mm, Ontario rejects wear of more than 2.0 mm when no OEM figure is available. Ontario's Passenger / Light-Duty Vehicle Inspection Standard rejects a shackle, pin or bushing when vertical movement of a spring or shackle against a spring pin exceeds the OEM standard or, if that is not available, when wear exceeds its own limits: 2.0 mm for a pin size up to 25 mm and 3.0 mm for a pin size greater than 25 mm. A 19 mm pin is in the smaller group. 3.0 mm is the limit for the larger pins, so applying it here would pass wear the standard rejects, and 4.0 mm is looser still. 1.0 mm is stricter than the standard. The same spring item separately rejects any leaf worn more than 3 mm in the hanger contact area or where leaves touch each other. Manitoba's handbook works differently, rejecting leaf-spring bushings or pivots worn in excess of 3 mm (1/8 in.), so the province decides which figure applies.

Q48. Answer: D

DOT brake fluid is hygroscopic — absorbs moisture from the atmosphere through brake system seals and hoses. Water contamination significantly reduces the wet boiling point of brake fluid and increases the risk of vapour lock during heavy braking. DOT 3: dry 205°C / wet 140°C. DOT 4: dry 230°C / wet 155°C. Moisture-laden fluid can boil during sustained heavy braking (mountain descents, track use), creating vapour bubbles that are compressible — resulting in a spongy or non-existent brake pedal (vapour lock). This is why brake fluid should be replaced every 2–3 years regardless of appearance.

Q49. Answer: D

Orange compressor oil means too much moisture has got into the oil. MAHLE's thermal management damage brochure says moisture causes orange discoloration, and the cause it gives is a filter-drier that was not changed when a component, for example the condenser, was replaced. The capacity of the drier's granules is limited, and moisture that enters the compressor oil reacts chemically with it, producing acids that attack nonferrous metals; the reaction between moisture and PAG oil is irreversible, and the vacuum of an air conditioning service unit cannot remove moisture from the oil. The system must be flushed thoroughly and the filter-drier and expansion valve replaced. In the same colour guide, too much contrast agent (leak-detecting dye) turns the oil green, metallic abrasion particles show in silvery, dark-green oil, and black, burned oil means the compressor got extremely hot.

Q50. Answer: D

The parking, tail, licence plate and side marker lamps. Section 108 of the Motor Vehicle Safety Regulations (CMVSS 108) requires the parking lamps, tail lamps, licence plate lamps and side marker lamps on a passenger car, multi-purpose passenger vehicle, three-wheeled vehicle, truck or bus to be activated while the front fog lamps are activated in a steady-burning state other than as daytime running lamps. The low beams are not in that list. The DRLs go the other way: beginning on September 1, 2020, they must be deactivated while the front fog lamps provide road illumination after being activated by the driver. Fog lamps that light with nothing else on do not meet the requirement. A 2024 vehicle is built after the section's August 31, 2019 transitional date, so the current text applies.

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